3D Multi-Pendulum Model of Slosh Dynamics in A Spacecraft

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***Abstract*—This paper presents a three-dimensional model for maneuvering of a spacecraft with unactuated fuel slosh dynamics, and uses Lagrangian for multi-pendulum analogy to obtain a mathematical model. Using of multi-pendulum model could be the best way for characterize the most fluid sloshing modes. Also, whole the fuel in the container is modeled by a fixed mass and two sloshing masses.**

Keywords*—*fuel slosh, mathematical model, three-dimensional model

I. INTRODUCTION

The satellites that work in higher orbits (like geosynchronous satellites), to arrive their orbits need to maneuver in space. Therefore, part of the mass of the spacecraft is allocated to fuel. If acceleration of the spacecraft changes, then, the fuel in the container will be affected with sloshing. This problem can disturb attitude of the spacecraft in orbital maneuvering [1,2] and might fail the mission. So, fuel sloshing is one of the important problems in attitude control.

The attitude control is one of the most important part of control system of spacecraft. Many studies have been done on this filed including mathematical modeling [3-5] and attitude control [6,7]. Some studies have been done on the interaction between spacecraft dynamics and fluid slosh dynamics, and they have demonstrated that pendulum and mass-spring models can approximate fluid dynamics. Previous works have used single pendulum model [8,9] or single mass-spring model [9] for characterization of lowest frequency sloshing mode, and recent works use multi-pendulum analogy [10] or multi-mass-spring analogy [11,12] for characterization of the most prominent sloshing modes, but all of this studies have been considered for planar maneuvering. In this paper, a spacecraft with a partially filled spherical fuel container that maneuver in three-dimensional space is considered, and multi-pendulum analogy have been used for characterization of the most prominent sloshing modes. In this model, pendulums can move in three dimensions. The torque M, two gimbal deflection angles δ_1, δ_2 and thrust (F) that act on center of mass of the spacecraft along the spacecraft's longitudinal axis (x-axis), y and z-axis are shown in Fig.1. This model can be employed to control attitude of the spacecraft in three-dimensional maneuvering.

II. MODELING

In this section, dynamics of the spacecraft with fuel slosh dynamics which maneuver in three-dimensional is formulated. Mass of the spacecraft is modeled by a rigid body (m) with moment of inertia (I), that they are constants and the rigid body is located on the center of mass of the spacecraft.

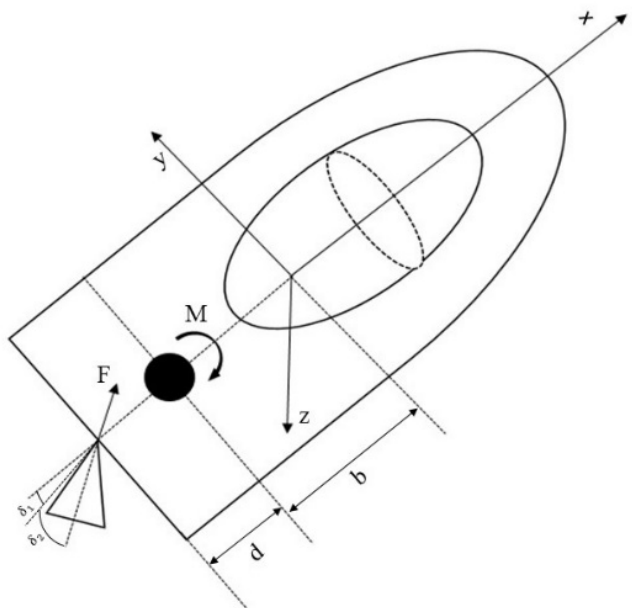

Fig.1 A spacecraft with a fuel container

The fuel container is spherical and fluid is modeled by rigidly attached mass (m_0) with moment of inertia (I_0) and pendulum masses m_i ($i=1,2,....,N$) with moment of inertia (I_i).

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The distances between unturblanced fuel, location of m_0 and location of m_i are $h_0 < 0$ and $h_i > 0$, respectively. Also, the distances between center of mass of the spacecraft, pivot gimbal and center of unturblanced fuel are d and b, respectively. Geometric characteristics of container, kind of fuel, and fill ratio of fuel container are effective factors on parameters of the multi-pendulum model (m_0, I_0, h_0, m_i, I_i, h_i).

Let $V \epsilon R^3$, $\Omega \epsilon R^3$, $\zeta \epsilon R^3$, $\tau_t \epsilon R^3$, $\tau_r \epsilon R^3$ denote the rigid body (m) translational velocity vector, the rigid body (m) angular velocity vector, and vector of pendulum masses coordinates, vector of generalized forces that act on rigid body (m) , vector of generalized moments that act on rigid body (m), respectively. Then equations of the spacecraft motion can derive from Kirchoff's equations [13] and extension of Lagrangian mechanic to include non-conservative forces as:

$$\frac{d}{dt}\frac{\partial L}{\partial V} + \bar{\Omega} \times \frac{\partial L}{\partial V} = \tau_t \tag{1}$$

$$\frac{d}{dt}\frac{\partial L}{\partial \Omega} + \bar{\Omega} \times \frac{\partial L}{\partial \Omega} + \bar{V} \times \frac{\partial L}{\partial V} = \tau_r \tag{2}$$

$$\frac{d}{dt}\frac{\partial L}{\partial \dot{\zeta}} - \frac{\partial L}{\partial \zeta} + \frac{\partial R}{\partial \dot{\zeta}} = 0 \tag{3}$$

where for a given vector $P = [p_1\ p_2\ p_3]^T \epsilon\ R^3$, $\bar{p}$ is skew-symmetric matrix, $\vartheta = [\vartheta_1\ \vartheta_2\ \vartheta_3]^T$ is position angles of m, $\zeta = [\varphi\ \ \theta\ \ \psi]^T$ is vector of position angles of m_i.

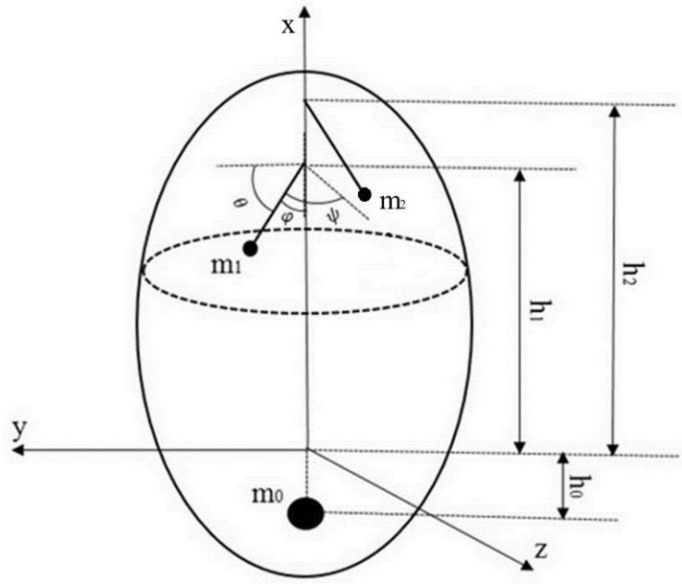

Fig.2 A three-dimensional multi pendulum model for fuel sloshing

Here, Lagrangian is expressed in terms of $V, \Omega, \zeta, \dot{\zeta}$ in other word $L = L(V, \Omega, \zeta, \dot{\zeta})$. also, R is Rayleigh dissipation function

$$R = \sum_{i=1}^{N} \varepsilon_i \left({\dot{\varphi}_i}^2 + {\dot{\theta}_i}^2 + {\dot{\psi}_i}^2\right)$$

In R, ε_i is damping constant for i-th mass.

As seen in Fig.1 and Fig.2 the position vectors of center of mass of the spacecraft (m), and fuel masses m_0, m_i, respectively, can be written as:

$r_c = [x\text{-}b\ \ y\ \ z]^T$, $r_0 = [x\text{-}h_0\ \ y\ \ z]^T$, $r_i = [x+h_i\text{-}l_i\cos\varphi_i\ \ y+l_i\cos\theta_i\ \ z+l_i\cos\psi_i]^T$

where l_i is length of i-th pendulum. Then we can compute velocity vectors as $V = \dot{r} + \Omega \times r$ for center of mass of the spacecraft and fuel masses. Where $\Omega = [\Omega_1\ \Omega_2\ \Omega_3]^T$ is angular velocity vector of center of mass of the spacecraft.

The potential energy is zero, because we assume that the spacecraft is in zero gravity environment. Thus the Lagrangian is equal to total kinetic energy ($L = T$), so we write:

$$L = \frac{1}{2} m {V_c}^2 + \frac{1}{2} m_0 {V_0}^2 + \frac{1}{2}\Omega^T(\mathrm{J} + I_0 \mathrm{I})\Omega + \frac{1}{2}\sum_{i=1}^{N}[m_i {V_i}^2 + (\omega_i + \Omega)^T(I_i \mathrm{I})(\omega_i + \Omega)]$$

Where $\mathrm{J} = \mathrm{diag}\{J_x\ \ J_y\ \ J_z\}$, and J_x, J_y, J_z are moment of inertia of the spacecraft along x, y and z-axis, respectively, and I is 3×3 identity matrix.

Applying equation 1-3 with

$$\omega_i = \begin{bmatrix} \dot{\varphi}_i \\ \dot{\theta}_i \\ \dot{\psi}_i \end{bmatrix}, V = \begin{bmatrix} v_x \\ v_y \\ v_z \end{bmatrix}, \tau_t = \begin{bmatrix} F\cos\delta_1 \cos\delta_2 \\ F\sin\delta_2 \\ -F\sin\delta_1 \cos\delta_2 \end{bmatrix},$$

$$\tau_r = \begin{bmatrix} M_x \\ M_y - F(b+d)\sin\delta_1 \cos\delta_2 \\ M_z - F(b+d)\sin\delta_2 \end{bmatrix}$$

Where ω_i is angular velocity of the pendulum masses, then, the equations of motion can be computed as

$$\begin{aligned}&(m + m_f)a_x + (mb + m_0 h_0)({\Omega_3}^2 + {\Omega_2}^2) + \\ &\sum_{i=1}^{2} m_i ({\Omega_2}^2 - {\Omega_3}^2)(h_i - l_i\cos\varphi_i) + \\ &\sum_{i=1}^{2} m_i l_i [{\dot{\varphi}_i}^2 \cos\varphi_i + 2\dot{\theta}_i \Omega_3 \sin\theta_i - 2\dot{\psi}_i \Omega_2 \sin\psi_i + \\ &\ddot{\varphi}_i \sin\varphi_i + \dot{\Omega}_2 \cos\psi_i - \dot{\Omega}_3 \cos\theta_i + \Omega_1\Omega_3 \cos\psi_i + \\ &\Omega_1\Omega_2 \cos\theta_i] = F\cos\delta_1 \cos\delta_2\end{aligned} \tag{4}$$

$$\begin{aligned}&(m + m_f)a_y - (mb + m_0 h_0)(\dot{\Omega}_3 + \Omega_1\Omega_2) + \\ &\sum_{i=1}^{2} m_i (\Omega_1\Omega_2 + \dot{\Omega}_3)(h_i - l_i\cos\varphi_i) + \\ &\sum_{i=1}^{2} m_i l_i \left[-{\dot{\theta}_i}^2 \cos\theta_i + 2\dot{\psi}_i \Omega_1 \sin\psi_i + 2\dot{\varphi}_i \Omega_3 \sin\varphi_i - \right. \\ &\left. \ddot{\theta}_i \sin\theta_i - \dot{\Omega}_1 \cos\psi_i - ({\Omega_1}^2 + {\Omega_3}^2)\cos\theta_i + \Omega_2\Omega_3\cos\psi_i\right] = \\ &F\sin\delta_2\end{aligned} \tag{5}$$

$$\begin{aligned}&(m + m_f)a_z - (mb + m_0 h_0)(\dot{\Omega}_2 - \Omega_1\Omega_3) + \\ &\sum_{i=1}^{2} m_i (\Omega_1\Omega_3 - \dot{\Omega}_2)(h_i - l_i\cos\varphi_i) + \\ &\sum_{i=1}^{2} m_i l_i \left[-{\dot{\psi}_i}^2 \cos\psi_i - 2\dot{\theta}_i \Omega_1 \sin\theta_i - 2\dot{\varphi}_i \Omega_2 \sin\varphi_i - \right.\end{aligned}$$

$$\ddot{\psi}_i sin\psi_i + \dot{\Omega}_1 cos\theta_i - \left({\Omega_1}^2 + {\Omega_2}^2\right)cos\psi_i + \Omega_1\Omega_2 cos\theta_i\Big] = -Fsin\delta_1 cos\delta_2 \tag{6}$$

$$\sum_{i=1}^{2} m_i {l_i}^2 \Big[\ddot{\theta}_i sin\theta_i cos\psi_i - \ddot{\psi}_i cos\theta_i sin\psi_i - 2\dot{\varphi}_i\Omega_3 sin\varphi_i sin\psi_i - 2\dot{\varphi}_i\Omega_2 \sin\varphi_i cos\theta_i + {\dot{\theta}_i}^2 cos\theta_i cos\psi_i - 2\dot{\theta}_i\Omega_1 sin\theta_i cos\theta_i - 2\dot{\psi}_i\Omega_1 sin\psi_i cos\psi_i - {\dot{\psi}_i}^2 cos\theta_i cos\psi_i + \dot{\Omega}_1 {cos\theta_i}^2 - \Omega_2\Omega_3 {cos\psi_i}^2 + \left({\Omega_3}^2 - {\Omega_2}^2\right)cos\theta_i cos\psi_i + \Omega_2\Omega_3 {cos\theta_i}^2\Big] + \sum_{i=1}^{2} m_i l_i \left[-\dot{\Omega}_3(h_i - l_i cos\varphi_i)cos\psi_i - \dot{\Omega}_2(h_i - l_i cos\varphi_i)cos\theta_i + \Omega_1\Omega_3(h_i - l_i cos\varphi_i)cos\theta_i - \Omega_1\Omega_2(h_i - l_i cos\varphi_i)cos\psi_i + a_z cos\theta_i - a_y cos\psi_i\right] + \sum_{i=1}^{2} m_i \left[\Omega_3 v_x(h_i - l_i cos\varphi_i) - \Omega_3 v_z(h_i - l_i cos\varphi_i) + \Omega_2\Omega_3(h_i - l_i cos\varphi_i)^2 - \Omega_2\Omega_3(h_i - l_i cos\varphi_i) + I_{xi}\left(\dot{\Omega}_1 + \ddot{\varphi}_i\right) - I_{yi}\left(\Omega_2\Omega_3 + \Omega_3\dot{\theta}_i\right) + I_{zi}\left(\Omega_2\Omega_3 + \Omega_2\dot{\psi}_i\right)\right] + \dot{\Omega}_1(J_x + I_0) + \Omega_2\Omega_3\left(J_z - J_y\right) = M_x \tag{7}$$

$$\sum_{i=1}^{2} m_i {l_i}^2 \left[\ddot{\varphi}_i sin\varphi_i cos\psi_i + {\dot{\varphi}_i}^2 cos\varphi_i cos\psi_i + \dot{\theta}_i\Omega_3 sin\theta_i cos\psi_i - 2\dot{\psi}_i\Omega_2 sin\psi_i cos\psi_i + \dot{\Omega}_2 {cos\psi_i}^2 - \dot{\Omega}_3 cos\theta_i cos\psi_i + \Omega_1\Omega_3 {cos\psi_i}^2 + \Omega_1\Omega_2 cos\theta_i cos\psi_i\right] + \sum_{i=1}^{2} m_i l_i \Big[\ddot{\psi}_i(h_i - l_i cos\varphi_i)sin\psi_i + {\dot{\psi}_i}^2(h_i - l_i cos\varphi_i)cos\psi_i + 2\dot{\theta}_i\Omega_1(h_i - l_i cos\varphi_i)sin\theta_i + \dot{\theta}_i\Omega_3 sin\theta_i cos\psi_i + 2\dot{\varphi}_i\Omega_2(h_i - l_i cos\varphi_i)sin\varphi_i - \dot{\Omega}_1(h_i - l_i cos\varphi_i)cos\theta_i + \left({\Omega_1}^2 - {\Omega_3}^2\right)(h_i - l_i cos\varphi_i)cos\psi_i - \Omega_2\Omega_3(h_i - l_i cos\varphi_i)cos\theta_i + a_x\Big] + \sum_{i=1}^{2} m_i \left[\left(\dot{\Omega}_2 - \Omega_1\Omega_3\right)(h_i - l_i cos\varphi_i)^2 - a_z(h_i - l_i cos\varphi_i) + I_{xi}(\Omega_1\Omega_3 + \dot{\varphi}_i\Omega_3) + I_{yi}\left(\ddot{\theta}_i + \dot{\Omega}_2\right) - I_{zi}\left(\Omega_1\Omega_3 + \dot{\psi}_i\Omega_3\right)\right] + \left(mb^2 + m_0{h_0}^2\right)\left(\dot{\Omega}_2 - \Omega_1\Omega_3\right) + (mb + m_0h_0)a_z + \dot{\Omega}_2\left(J_y + I_0\right) + \Omega_1\Omega_3\left(J_x - J_z\right) = M_y - F(b+d)sin\delta_1 cos\delta_2 \tag{8}$$

$$\sum_{i=1}^{2} m_i {l_i}^2 \Big[-\ddot{\varphi}_i sin\varphi_i cos\theta_i - \ddot{\theta}_i(h_i - l_i cos\varphi_i)sin\theta_i - {\dot{\varphi}_i}^2 cos\varphi_i cos\theta_i - 2\dot{\theta}_i\Omega_3 sin\theta_i cos\theta_i + 2\dot{\theta}_i\Omega_2 sin\theta_i cos\psi_i + 2\dot{\psi}_i\Omega_2 cos\theta_i sin\psi_i + \dot{\Omega}_3 {cos\theta_i}^2 - \dot{\Omega}_2 cos\theta_i cos\psi_i - \Omega_1\Omega_2 {cos\theta_i}^2 - \Omega_1\Omega_3 cos\theta_i cos\psi_i\Big] + \sum_{i=1}^{2} m_i l_i \Big[-{\dot{\theta}_i}^2(h_i - l_i cos\varphi_i)cos\theta_i + 2\dot{\varphi}_i\Omega_3(h_i - l_i cos\varphi_i)sin\varphi_i + 2\dot{\psi}_i\Omega_1(h_i - l_i cos\varphi_i)sin\psi_i - \dot{\Omega}_1(h_i - l_i cos\varphi_i)cos\psi_i + \left({\Omega_2}^2 - {\Omega_1}^2\right)(h_i - l_i cos\varphi_i)cos\theta_i - \Omega_2\Omega_3(h_i - l_i cos\varphi_i)cos\psi_i - a_x cos\theta_i\Big] + \sum_{i=1}^{2} m_i \left[\left(\dot{\Omega}_3\Omega_1\Omega_2\right)(h_i - l_i cos\varphi_i)^2 + a_y(h_i - l_i cos\varphi_i) + I_{xi}(\Omega_1\Omega_2 + \dot{\varphi}_i\Omega_2) + I_{zi}\left(\ddot{\psi}_i + \dot{\Omega}_3\right) + I_{yi}\left(\Omega_1\Omega_2 + \dot{\theta}_i\Omega_1\right)\right] + \left(mb^2 + m_0{h_0}^2\right)\left(\dot{\Omega}_3 + \Omega_1\Omega_2\right) - (mb + m_0h_0)a_y + \dot{\Omega}_3(J_z + I_0) + \Omega_1\Omega_3\left(J_y - J_x\right) = M_z - F(b+d)sin\delta_2 \tag{9}$$

$$\sum_{i=1}^{2} m_i {l_i}^2 \left[\ddot{\varphi}_i {sin\varphi_i}^2 + {\dot{\varphi}_i}^2 sin\varphi_i cos\varphi_i + 2\dot{\theta}_i\Omega_3 sin\varphi_i sin\theta_i - 2\dot{\psi}_i\Omega_2 sin\varphi_i sin\psi_i + \dot{\Omega}_2 sin\varphi_i cos\psi_i - \dot{\Omega}_3 sin\varphi_i cos\theta_i + \Omega_1\Omega_2 sin\varphi_i cos\psi_i\right] + \sum_{i=1}^{2} m_i l_i \left[-\left({\Omega_2}^2 + {\Omega_3}^2\right)(h_i - l_i cos\varphi_i)sin\varphi_i + a_x sin\varphi_i\right] + \sum_{i=1}^{2} m_i I_{xi}\left(\dot{\Omega}_1 + \ddot{\varphi}_i\right) + \epsilon_i\dot{\varphi}_i = 0 \tag{10}$$

$$\sum_{i=1}^{2} m_i {l_i}^2 \Big[\ddot{\theta}_i {sin\theta_i}^2 + {\dot{\theta}_i}^2 sin\theta_i cos\theta_i - 2\dot{\varphi}_i\Omega_3 sin\varphi_i sin\theta_i - 2\dot{\psi}_i\Omega_1 sin\theta_i sin\psi_i + \dot{\Omega}_1 sin\theta_i cos\psi_i - \Omega_2\Omega_3 sin\theta_i cos\psi_i + \left({\Omega_1}^2 + {\Omega_3}^2\right)\sin\theta_i cos\theta_i\Big] + \sum_{i=1}^{2} m_i l_i \left[-(\dot{\Omega}_3 + \Omega_1\Omega_2)(h_i - l_i cos\varphi_i)sin\theta_i - a_y sin\theta_i\right] + \sum_{i=1}^{2} m_i I_{yi}\left(\dot{\Omega}_2 + \ddot{\theta}_i\right) + \epsilon_i\dot{\theta}_i = 0 \tag{11}$$

$$\sum_{i=1}^{2} m_i {l_i}^2 \Big[\ddot{\psi}_i {sin\psi_i}^2 + {\dot{\psi}_i}^2 sin\psi_i cos\psi_i - 2\dot{\varphi}_i\Omega_2 sin\varphi_i sin\psi_i + 2\dot{\theta}_i\Omega_1 sin\theta_i sin\psi_i - \dot{\Omega}_1 sin\psi_i cos\theta_i - \Omega_2\Omega_3 sin\psi_i cos\theta_i + \left({\Omega_1}^2 + {\Omega_2}^2\right)\sin\psi_i cos\psi_i\Big] + \sum_{i=1}^{2} m_i l_i \left[(\dot{\Omega}_2 + \Omega_1\Omega_3)(h_i - l_i cos\varphi_i)sin\psi_i - a_z sin\psi_i\right] + \sum_{i=1}^{2} m_i I_{zi}\left(\dot{\Omega}_3 + \ddot{\psi}_i\right) + \epsilon_i\dot{\psi}_i = 0 \tag{12}$$

Where I_{xi}, I_{yi}, I_{zi} are moments of inertia of i-th mass along x, y and z-axis, respectively, $m_0 + \sum_{i=1}^{N} m_i = m_f$ is sum of all the fuel masses, and $a = [a_x\ a_y\ \ a_y]^T = \dot{V} + \Omega \times V$ is the acceleration of the center of mass of the fuel in spacecraft frame. For prove this modeling, if assume that attitude of the spacecraft can only change about y-axis, and ignore the force that F creates along z-axis and torques M_x, M_z, also, if the sloshing masses can only move in xy-plane and ignore their angles relative to the z-axis, then, this modeling changes to a two-dimensional fuel sloshing model, which is modeled by multi-pendulum analogy, and this is equivalent to the model that is described in [10].

III. CONCLUSION

In this paper, we developed a three-dimensional mathematical model for a spacecraft with sloshing modes. For the first time, this paper use multi pendulum model in three-dimensional space that it can approximate complicated fluid motion in spacecraft container very well. Our Future researches includes design a feedback control for this new modeling of problem.

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